1. lsof – List Open Files

Displays files currently opened by processes.

```
vanam@Uday:~$ sudo lsof -i :8080
[sudo] password for vanam:
COMMAND PID    USER   FD   TYPE DEVICE SIZE/OFF NODE NAME
java    185 jenkins    9u  IPv6   7471      0t0  TCP *:http-alt (LISTEN)
vanam@Uday:~$
```

2. fuser – Identify Processes Using Files or Ports

```
vanam@Uday:~$ sudo fuser -v 8080/tcp
                     USER        PID ACCESS COMMAND
8080/tcp:            jenkins     185 F.... java
vanam@Uday:~$
```

3. du – Check Directory Size

- `-s` – Display only the total.
- `-h` – Human-readable format (KB, MB, GB).

```
vanam@Uday:~$ sudo du -sh /var/log
427M    /var/log
vanam@Uday:~$
```

4. journalctl – View System Logs

```
-- No entries --
vanam@Uday:~$ sudo journalctl -u jenkins --since "1 month ago"
Oct 21 12:35:59 Uday systemd[1]: Starting jenkins.service - Jenkins Continuous Integration Server...
Oct 21 12:35:59 Uday jenkins[2277]: Running from: /usr/share/java/jenkins.war
Oct 21 12:35:59 Uday jenkins[2277]: 2026-10-21 12:35:59.437+0000 [id=1]        INFO        winstone.Logger#logInternal: Beginning extraction from war file
Oct 21 12:35:59 Uday jenkins[2277]: 2026-10-21 12:35:59.875+0000 [id=1]        WARNING        o.e.j.ee9.nested.ContextHandler#setContextPath: Empty context>
Oct 21 12:35:59 Uday jenkins[2277]: 2026-10-21 12:35:59.934+0000 [id=1]        INFO        org.eclipse.jetty.server.Server#doStart: jetty-12.1.13; built: 2>
Oct 21 12:36:00 Uday jenkins[2277]: 2026-10-21 12:36:00.281+0000 [id=1]        INFO        o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Suppo>
Oct 21 12:36:00 Uday jenkins[2277]: 2026-10-21 12:36:00.324+0000 [id=1]        INFO        o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
Oct 21 12:36:00 Uday jenkins[2277]: 2026-10-21 12:36:00.629+0000 [id=1]        INFO        hudson.WebAppMain#contextInitialized: Jenkins home directory: /v>
Oct 21 12:36:00 Uday jenkins[2277]: 2026-10-21 12:36:00.729+0000 [id=1]        INFO        o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHa>
Oct 21 12:36:00 Uday jenkins[2277]: 2026-10-21 12:36:00.739+0000 [id=1]        INFO        o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConne>
Oct 21 12:36:00 Uday jenkins[2277]: 2026-10-21 12:36:00.740+0000 [id=1]        INFO        org.eclipse.jetty.server.Server#doStart: Started oejs.Server@5bc>
Oct 21 12:36:00 Uday jenkins[2277]: 2026-10-21 12:36:00.742+0000 [id=43]        INFO        winstone.Logger#logInternal: Winstone Servlet Engine running: c>
Oct 21 12:36:00 Uday jenkins[2277]: 2026-10-21 12:36:00.830+0000 [id=40]        INFO        jenkins.model.Jenkins#<init>: Starting version 2.580.1
Oct 21 12:36:00 Uday jenkins[2277]: 2026-10-21 12:36:00.962+0000 [id=51]        INFO        jenkins.InitReactorRunner$1#onAttained: Started initialization
Oct 21 12:36:00 Uday jenkins[2277]: 2026-10-21 12:36:00.970+0000 [id=72]        INFO        jenkins.InitReactorRunner$1#onAttained: Listed all plugins
Oct 21 12:36:01 Uday jenkins[2277]: 2026-10-21 12:36:01.714+0000 [id=72]        INFO        jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
Oct 21 12:36:01 Uday jenkins[2277]: 2026-10-21 12:36:01.718+0000 [id=72]        INFO        jenkins.InitReactorRunner$1#onAttained: Started all plugins
Oct 21 12:36:01 Uday jenkins[2277]: 2026-10-21 12:36:01.720+0000 [id=71]        INFO        jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
Oct 21 12:36:01 Uday jenkins[2277]: 2026-10-21 12:36:01.897+0000 [id=49]        INFO        jenkins.InitReactorRunner$1#onAttained: System config loaded
Oct 21 12:36:01 Uday jenkins[2277]: 2026-10-21 12:36:01.898+0000 [id=65]        INFO        jenkins.InitReactorRunner$1#onAttained: System config adapted
Oct 21 12:36:01 Uday jenkins[2277]: 2026-10-21 12:36:01.899+0000 [id=70]        INFO        jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
Oct 21 12:36:01 Uday jenkins[2277]: 2026-10-21 12:36:01.901+0000 [id=70]        INFO        jenkins.InitReactorRunner$1#onAttained: Configuration for all j>
Oct 21 12:36:01 Uday jenkins[2277]: 2026-10-21 12:36:01.924+0000 [id=88]        INFO        hudson.util.Retrier#start: Attempt #1 to do the action check up>
Oct 21 12:36:02 Uday jenkins[2277]: 2026-10-21 12:36:02.201+0000 [id=70]        INFO        jenkins.install.SetupWizard#init:
```

5.ss – Display Network Connections

```
vanam@Uday:~$ sudo ss -tulnp
Netid       State       Recv-Q      Send-Q           Local Address:Port             Peer Address:Port      Process
udp         UNCONN      0           0                   127.0.0.54:53                  0.0.0.0:*          users:(("systemd-resolve",pid=162,fd=16))
udp         UNCONN      0           0                127.0.0.53%lo:53                  0.0.0.0:*          users:(("systemd-resolve",pid=162,fd=14))
udp         UNCONN      0           0             10.255.255.254:53                    0.0.0.0:*
udp         UNCONN      0           0                  127.0.0.1:323                   0.0.0.0:*
udp         UNCONN      0           0                      [::1]:323                      [::]:*
tcp         LISTEN      0           4096               127.0.0.1:40849                 0.0.0.0:*          users:(("containerd",pid=272,fd=15))
tcp         LISTEN      0           4096                127.0.0.54:53                  0.0.0.0:*          users:(("systemd-resolve",pid=162,fd=17))
tcp         LISTEN      0           1000            10.255.255.254:53                  0.0.0.0:*
tcp         LISTEN      0           4096             127.0.0.53%lo:53                  0.0.0.0:*          users:(("systemd-resolve",pid=162,fd=15))
tcp         LISTEN      0           50                           *:8080                     *:*          users:(("java",pid=185,fd=9))
vanam@Uday:~$
```

6. timeout – Execute a Command for a Limited Time

```
vanam@Uday:~$ timeout 10s ping google.com
PING google.com (192.178.174.138) 56(84) bytes of data.
64 bytes from lcbomo-in-f138.1e100.net (192.178.174.138): icmp_seq=1 ttl=106 time=176 ms
64 bytes from lcbomo-in-f138.1e100.net (192.178.174.138): icmp_seq=2 ttl=106 time=32.1 ms
64 bytes from lcbomo-in-f138.1e100.net (192.178.174.138): icmp_seq=3 ttl=106 time=31.0 ms
64 bytes from lcbomo-in-f138.1e100.net (192.178.174.138): icmp_seq=4 ttl=106 time=31.3 ms
64 bytes from lcbomo-in-f138.1e100.net (192.178.174.138): icmp_seq=5 ttl=106 time=30.9 ms
64 bytes from lcbomo-in-f138.1e100.net (192.178.174.138): icmp_seq=6 ttl=106 time=30.8 ms
64 bytes from lcbomo-in-f138.1e100.net (192.178.174.138): icmp_seq=7 ttl=106 time=31.9 ms
64 bytes from lcbomo-in-f138.1e100.net (192.178.174.138): icmp_seq=8 ttl=106 time=31.3 ms
64 bytes from lcbomo-in-f138.1e100.net (192.178.174.138): icmp_seq=9 ttl=106 time=32.7 ms
64 bytes from lcbomo-in-f138.1e100.net (192.178.174.138): icmp_seq=10 ttl=106 time=31.0 ms
vanam@Uday:~$
```



7.watch – Monitor Command Output Continuously

```
Every 2.0s: free -h                                                    Uday: Thu Nov  5 18:58:12 2026

               total        used        free      shared  buff/cache   available
Mem:           7.6Gi       1.4Gi       5.3Gi       3.5Mi       1.1Gi       6.2Gi
Swap:          2.0Gi          0B       2.0Gi
```

8. stat – Display File Metadata

```
vanam@Uday:~$ stat /var/log/syslog
  File: /var/log/syslog
  Size: 41460           Blocks: 88         IO Block: 4096   regular file
Device: 8,48    Inode: 33556256    Links: 1
Access: (0640/-rw-r-----)  Uid: (  102/  syslog)   Gid: (    4/     adm)
Access: 2026-11-05 18:51:54.386057708 +0000
Modify: 2026-11-05 18:57:58.915971541 +0000
Change: 2026-11-05 18:57:58.915971541 +0000
 Birth: 2026-11-05 18:51:54.386057708 +0000
vanam@Uday:~$
```

9. find – Find Files Based on Modification Time

```
vanam@Uday:~$ sudo find /var/log -type f -mtime -1
/var/log/journal/cc850cc02efb43bba649b59cec803312/user-1000.journal
/var/log/journal/cc850cc02efb43bba649b59cec803312/system@598a1f69e2db46f7badcbc568a481ac1-0000000000006207-00065f8bfd2b9118.journal
/var/log/journal/cc850cc02efb43bba649b59cec803312/system.journal
/var/log/journal/cc850cc02efb43bba649b59cec803312/user-1000@42a86a6ce6b946949f64527cf4d7caf5-000000000000605b-00065f5000b3dcb6.journa
/var/log/dmesg
/var/log/wtmp
/var/log/syslog.1
/var/log/auth.log.1
/var/log/syslog
/var/log/kern.log
/var/log/auth.log
/var/log/kern.log.1
```

10. lscpu – Display CPU Information



```
/var/log/kern.log.1
vanam@Uday:~$ lscpu
Architecture:             x86_64
  CPU op-mode(s):         32-bit, 64-bit
  Address sizes:          39 bits physical, 48 bits virtual
  Byte Order:             Little Endian
CPU(s):                   12
  On-line CPU(s) list:    0-11
Vendor ID:                GenuineIntel
  Model name:             12th Gen Intel(R) Core(TM) i5-1235U
    CPU family:           6
    Model:                154
    Thread(s) per core:   2
    Core(s) per socket:   6
    Socket(s):            1
    Stepping:             4
    BogoMIPS:             4992.00
    Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp
                          lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1
                           sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp
                           ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha
                          _ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnmi umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_clear serialize
                           flush_l1d arch_capabilities
Virtualization features:
  Virtualization:         VT-x
  Hypervisor vendor:      Microsoft
  Virtualization type:    full
Caches (sum of all):
  L1d:                    288 KiB (6 instances)
  L1i:                    192 KiB (6 instances)
  L2:                     7.5 MiB (6 instances)
  L3:                     12 MiB (1 instance)
NUMA:
  NUMA node(s):           1
  NUMA node0 CPU(s):      0-11
Vulnerabilities:
  Gather data sampling:   Not affected
  Itlb multihit:          Not affected
  L1tf:                   Not affected
  Mds:                    Not affected
  Meltdown:               Not affected
  Mmio stale data:        Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed:               Mitigation; Enhanced IBRS
  Spec rstack overflow:   Not affected
  Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:             Mitigation; Enhanced / Automatic IBRS; IBPB conditional; RSB filling; PBRSB-eIBRS SW sequence; BHI BHI_DIS_S
  Srbds:                  Not affected
  Tsx async abort:        Not affected
```